

Intermediate Macro

Chapter 9: The $IS - LM - PC$ Model

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and the following results are obtained:

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- Chapter 9 ties *IS – LM* **model** and **Phillips curve** together to analyze the impact of a shock or policy on the economy.
- Chapter 9 introduces the concepts of **potential output**, actual output, and the **output gap**.
- Chapter 9 makes extensive use of **dynamic analysis**, introduces the term **business cycle**, and distinguishes between **shocks** and **propagation mechanism**.

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- Chapter 9 assumes that the expected price level adjusts to differences between the actual and (previously) expected price levels.
- If the actual price level is greater (less) than the expected price level, wage setters assumed to increase (decrease) their expected price level in the future.
- This adjustment mechanism is essential for the dynamic analysis.

Recalling the IS and LM Relation

9.1 The $IS - LM - PC$ Model

- Consider the $IS - LM$ diagram from Chapter 6 (equation 6.5).
- From *equilibrium in the goods market*, in the short run, output is determined by demand and given by the equation:

$$Y = C(Y - T) + I(Y, r + x) + G \quad (9.1)$$

- The IS curve represents these relationships graphically.
- From *equilibrium in financial markets*, we derived a flat LM curve, showing the real rate r chosen by the central bank.

Recalling the Phillips Curve

9.1 The $IS - LM - PC$ Model

- From *equilibrium in labor markets*, we derived the following equation (equation 8.10), a relation we called the *Phillips curve*:

$$\pi - \pi^e = -\alpha(u - u_n) \quad (9.2)$$

- Given that the (9.1) is in terms of output, we must express this Phillips curve relation **in terms of output and inflation** rather than unemployment and inflation.
- By definition, the unemployment rate is equal to unemployment divided by the labor force:

$$u \equiv \frac{U}{L} = \frac{L - N}{L} = 1 - \frac{N}{L}$$

where N denotes employment and L the labor force.

The Natural Rate of Unemployment and Potential Output

9.1 The $IS - LM - PC$ Model

- Reorganizing to express N as a function of u gives:

$$N = L(1 - u)$$

- Turning to output, we shall maintain for the moment the simplifying assumption we made in Chapter 7, that *output is equal to employment*:

$$Y = N = L(1 - u)$$

- Thus, when the unemployment rate is equal to the natural rate u_n , employment is given by $N_n = L(1 - u_n)$ and output is equal to $Y_n = L(1 - u_n)$.
- Y_n is called the natural level of output or also called **potential output**.

Output Gap

9.1 The $IS - LM - PC$ Model

- It follows that we can express the deviation of output from its natural level as:

$$Y - Y_n = L((1 - u) - (1 - u_n)) = -L(u - u_n)$$

- The difference between actual output and potential output is called the **output gap**.
- This gives us a simple relation between the deviation of output from potential and the deviation of unemployment from its natural rate.
- Recall that the relation of this equation to the actual relation between output and unemployment is known as Okun's law.

The Relation Between Inflation and Output

9.1 The $IS - LM - PC$ Model

- Replacing $u - u_n$ in equation (9.2) gives:

$$\pi - \pi^e = \frac{\alpha}{L}(Y - Y_n) \quad (9.3)$$

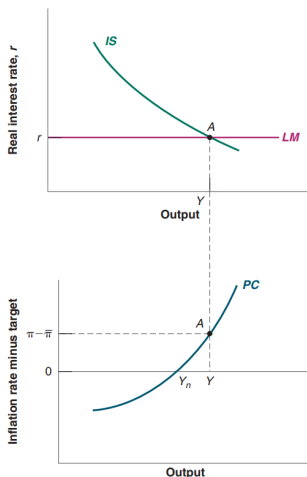
- We need to take one last step. We saw that in Chapter 8 that the way wage setters form **expectations** about inflation has changed through time.
- The evidence in Chapter 8 suggests that, in the United States today, inflation expectations are anchored and a reasonable assumption is that wage setters expect inflation to be equal to the target set by the Fed, denoted $\bar{\pi}$.
- With this assumption, the relation between inflation and output is given by:

$$\pi - \bar{\pi} = \frac{\alpha}{L}(Y - Y_n) \quad (9.4)$$

The $IS - LM - PC$ Model: Output and Inflation

9.1 The $IS - LM - PC$ Model

- The following figure shows the *short-run* equilibrium.



What Happens over Time?

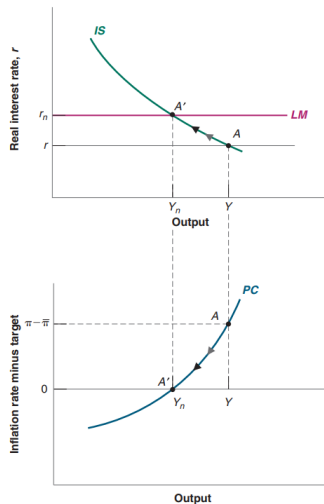
9.2 From the Short to the Medium Run

- In the situation of the previous figure, suppose the central bank leaves the real rate unchanged at r .
- Then, output remains above target. At some point, policy is likely to react to this higher inflation.
- The central bank will react to the positive output gap by increasing the real rate so as to reduce inflation, in so doing, reduce output.

Medium-Run Output and Inflation

9.2 From the Short to the Medium Run

- The process of adjustment is described in the following figure.



The Natural Rate of Interest

9.2 From the Short to the Medium Run

- At point A' , the economy reaches its *medium-run* equilibrium.
- The real rate must be such that the demand for goods is equal to potential output, and is thus given by $r = \bar{r}$. This interest rate is often called the **natural rate of interest**.
- It is also sometimes called the **neutral rate of interest** or the **Wicksellian rate of interest**.
- What about the nominal interest i ?
- Recall the relation between the nominal and the real rate from Chapter 6 (equation 6.4): $i = r + \pi^e$.
- Given that, in the *medium run*, the real rate is equal to the neutral rate, and that expected inflation is equal to actual inflation, which is itself equal to target inflation, this implies that the nominal rate is equal to $i = r_n + \bar{\pi}$.

Neutrality of Money

9.2 From the Short to the Medium Run

- Finally, what about money and money growth?
- Recall the characterization of the financial markets equilibrium condition in Chapter 5 (equation (5.3)) that $\frac{M}{P} = Y \times L(i)$.
- Given that output is equal to potential, and the nominal interest rate is determined as the previous slide, we can rewrite this equation as:

$$\frac{M}{P} = Y_n \times L(r_n + \bar{\pi})$$

- This implies that the price level P must be growing at the same rate as nominal money M ($\pi = g_M$ where g_M denotes the rate of money growth).
- To summarize, in the *medium run*, the real variables, such as output, unemployment, and the real rate of interest, are ***independent of monetary policy***.

Theory versus Reality

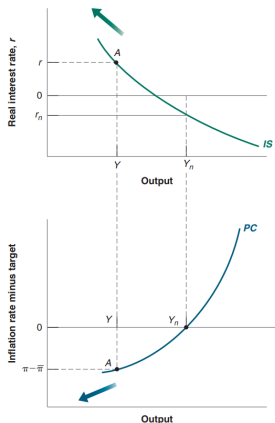
9.3 Complications and How Things Can Go Wrong

- The central bank adjustment process seems simple in theory but is more complex in reality.
- One factor that complicates matters is the difficulty in determining where potential output is exactly and thus how far current output is from potential.
- For this reason, central banks may adjust the real interest rate slowly to see what happens.
- It also takes time for the economy to respond to policy changes, so a slower approach is typically adopted.

A Recession Case

9.3 Complications and How Things Can Go Wrong

- In the previous section, we considered the case where output was above potential and inflation was above target.
- Consider instead the case where the economy is in a recession.



The Zero Lower Bound and Deflation Spirals

9.3 Complications and How Things Can Go Wrong

- What the central bank should do in this case appears straightforward: It should decrease the real rate until output has increased back to its natural level.
- The zero lower bound may, however, make it impossible to achieve this negative real rate because the lowest nominal rate monetary policy can achieve is a rate of 0%.
- When the economy is zero lower bound, the central bank may not be able to lower policy rates to stimulate output.
- Even lowering interest rates to negative levels may not work and could result in continued falling output and prices, known as **deflation trap** or **deflation spiral**.
- This scenario occurred in the Great Depression from 1929 to 1933.

Medium-run Effects of Changes in Fiscal Policy

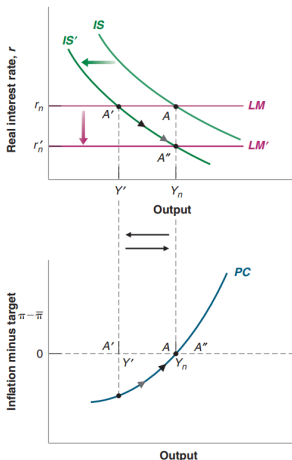
9.4 Fiscal Consolidation Revisited

- We can now use $IS - LM - PC$ model to determine the medium-run effects of changes in fiscal policy.
- Suppose that output is at potential.
- Now, assume that the government attempts deficit reduction via higher taxes.
- This reduces consumption and shifts the IS curve to the left. This shift results in lower output which, in turn, lowers investment.
- Output will now be below the potential level of output.
- Monetary action may now be needed to increase output back to the potential level of output.
- All that is needed is for the central bank and the government to coordinate carefully.

Fiscal Consolidation in the Short and the Medium Run

9.4 Fiscal Consolidation Revisited

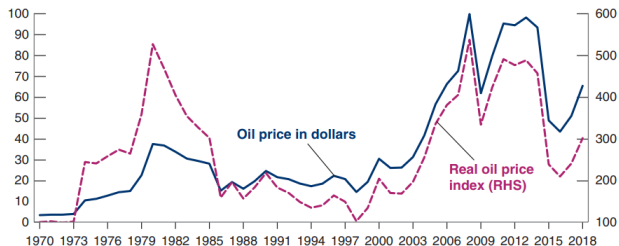
- Monetary and fiscal policy need to work *in tandem* to keep the economy at the potential level of output.



Shifts in PC curve

9.5 The Effects of an Increase in the Price of Oil

- So far, we have looked at shocks to demand, shocks that shifted the IS curve but left potential output and thus the position of the PC curve unaffected.
- There are other shocks, however, that affect both demand and potential output and play an important role in fluctuations.
- An obvious candidate is movements in the price of oil.



Including Oil Price into the Model

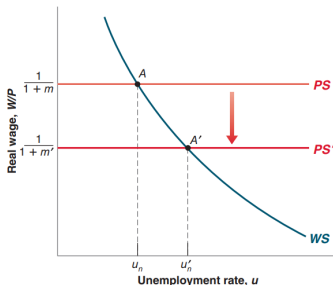
9.5 The Effects of an Increase in the Price of Oil

- Until now, we have assumed that output was produced using only labor. One way to extend our model would be to recognize explicitly that output is produced using labor *and* other inputs (including *energy*).
- An easier way is simply to capture the increase in the price of oil by an increase in m —the markup of the price over the nominal wage.
- Having made this assumption, we can then track of the dynamic effects of an *increase in the markup* on output and inflation.

Effects on the Natural Rate of Unemployment

9.5 The Effects of an Increase in the Price of Oil

- We need to return to Chapter 7 and the characterization of labor market equilibrium.
- The wage-setting relation (WS) is downward sloping: a higher unemployment rate leads to lower real wages.
- The price-setting relation (PS) is represented by the horizontal line at $\frac{W}{P} = \frac{1}{1+m}$.

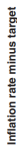


Effects on the Natural Level of Output

9.5 The Effects of an Increase in the Price of Oil

- The increase in the natural rate of unemployment leads in turn to a decrease in the natural level of employment.
- If we assume that the relation between employment and output is unchanged, then the decrease in the natural level of employment leads to an identical decrease in potential output.

9.5 The Effects of an Increase in the Price of Oil



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The Short Run versus the Medium Run

9.6 Conclusions

- One key message of this chapter is that shocks or changes in policy typically have different effects in the short run and in the medium run.
- Disagreements among economists about the effects of various policies often come from differences in the time frame they have in mind.
- One implication is that where you stand depends on how fast you think the economy adjusts to shocks.
- Typically, the economy returns over time to its medium-run equilibrium. A major issue is whether we can expect the economy to return to that equilibrium on its own, or whether it requires an active policy response.

Shocks and Propagation Mechanisms

9.6 Conclusions

- This chapter also gives you a general way of thinking about **output fluctuations** (sometimes called **business cycles**)—movements in output around its trend.
- You can think of the economy as being constantly hit by **shocks**.
- Each shock has dynamic effects on output and its components. These dynamic effects are called the **propagation mechanism** of the shock. Propagation mechanisms are different for different shocks.